Wine Quality Prediction

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Abstract

This data analysis was to predict the wine quality based on physicochemical properties and wine preferences. The data set collected are separated by red and white wine. Each individual data set includes measurements on physicochemical properties and the corresponding sensory data. In this report, we present the models where we applied linear regression and generalized additive models to find the best prediction for each type of wine's quality. We will use AIC as an error metric, lower AIC yields a better model.

Keywords: wine quality, multiple linear regression, generalized additive models

1 Introduction

Wine and food are known to be an important part of the French culture as well as many other European countries. Vinho Verde is recognized as the first Portuguese wines exported to European markets, and it has recurrently been awarded in national and international wine competitions (Petropoulos et al. 2017). Being innovative and considerate about their wine quality, the company wishes to uncover any relation between physicochemical properties and wine preferences so as to predict wine quality base on those properties. The data set collected are separated by red and white wine. Each individual data set includes measurements on physicochemical properties and the corresponding sensory data.

In this report, we present the models where we applied linear regression and generalized additive models to find the best prediction for each type of wine's quality. We will use AIC as an error metric, lower AIC yields a better model.

2 Background

In the data sets, there are a total of 1599 red wine samples and 4898 white wine samples, with none of unavailable data within each. The physicochemical properties of the wine samples were collected in form of objective tests, and the sensory data was the median of at least three evaluations made by wine experts. The grade given to each sample is on the scale from 0 to 10, with 0 being very bad and 10 being very excellent. However, the range of that for red wine samples is from 3 to 8 and 3to 9 for white wine samples and their distributions can be seen from Figure 1.

For a general look of the physicochemical properties of each type of wine, Table 1 (Dogucu & Çetinkaya-Rundel 2020) lists the data statistics of the variables. At first glance, white wine contains more of residual sugar, free sulfur dioxide and total sulfur dioxide due to the larger range and higher mean of these variables from white wine samples. In addition, from the boxplots of the variables, they all contains outliers except for alcohol.

Now, consider the correlation between the variables in each data set. Observing from the correlation matrix of red wine data set in Figure 2, free sulfur dioxide is positively correlated to total sulfur dioxide, same as fixed acidity and density. Other correlations

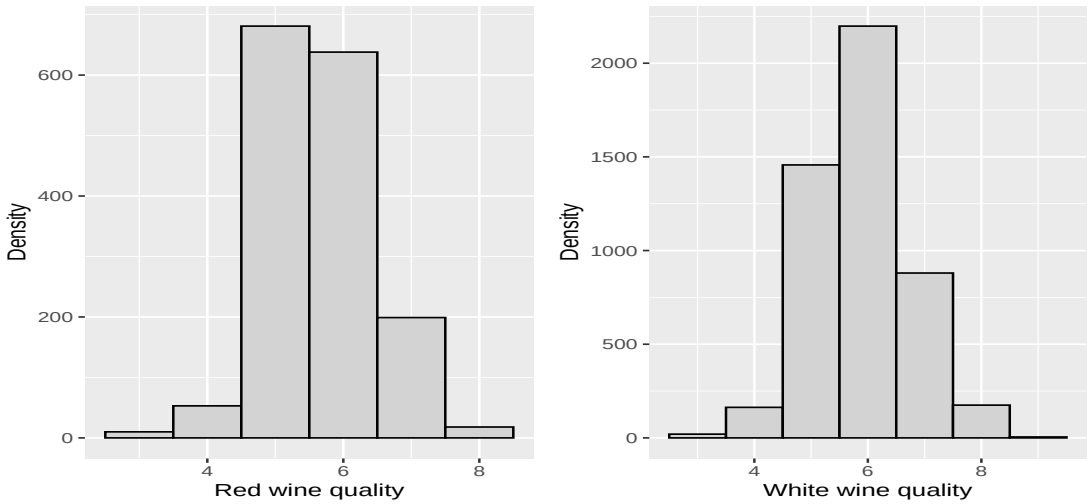

Figure 1: Histograms of wine quality

Properties	Min	Mean	Max
Fixed acidity	4.60	8.32	15.90
Vovatile acidity	0.12	0.53	1.58
Citric acid	0.00	0.27	1.00
Residual sugar	0.90	2.54	15.50
Chlorides	0.01	0.09	0.61
Free sulfur dioxide	1.00	15.87	72.00
Total sulfur dioxide	6.00	46.47	289.00
Density	0.99	1.00	1.00
pH	2.74	3.31	4.01
Sulphates	0.33	0.66	2.00
Alcohol	8.40	10.42	14.90

Table 1: Univariate statistics of red wine

can be seen from the matrix to aid the process of model selection, especially, the linear relationship between the physicochemical variables and wine quality are indicated from the matrix. The white wine data set is analyzed in the same manner for correlations of the variables (Hu et al. 2016).

However, plotting the variable itself with wine quality, the linear correlation does not

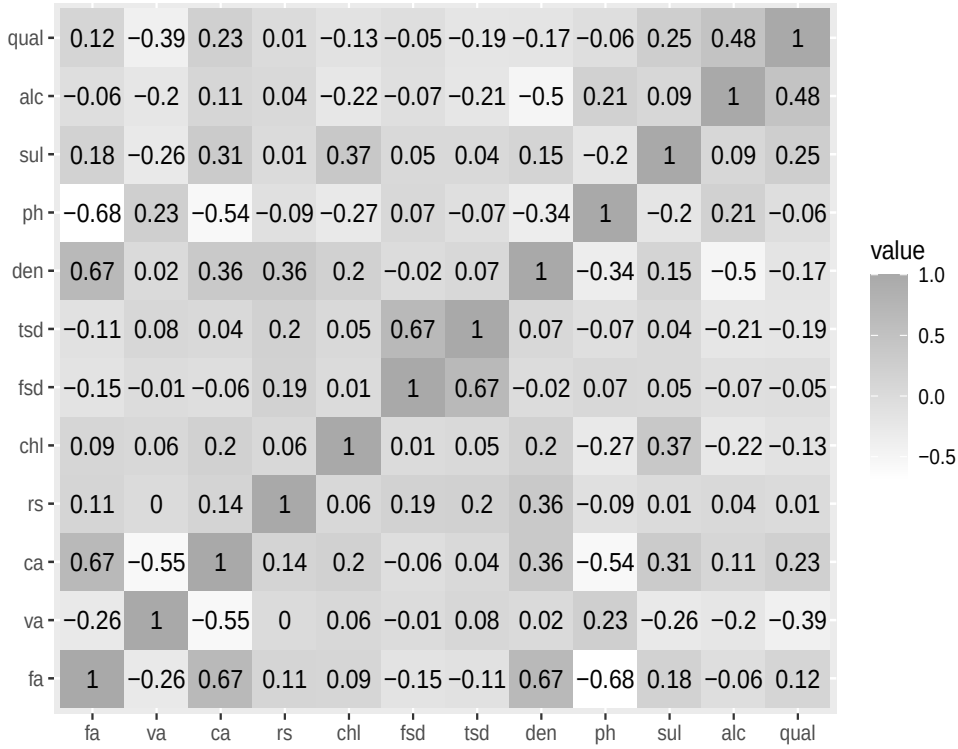

Figure 2: Correlation between the variables in red wine data set

correctly represent the relationship between that variable and quality. For examples, in Figure 3, the plot on the left is from the red wine data of sulphates versus wine quality, we can see that the linear line does not fit the data appropriately. Similarly, the plot on the right from the white wine data set of chlorides versus wine quality indicates the same issue. Thus, nonlinear models should also be fitted to help produce a better prediction for wine quality.

3 Methods

3.1 Red Wine Data Set

We first attempt to fit a Multiple Linear Regression model with all the variables available. From the output, there are insignificant predictors present in the model on the basis of p-values, thus, we proceed with applying Best Subset Selections. The outputs are displayed in Table 2. On the basis of AIC, the model with 7 predictors has the lowest AIC, call it

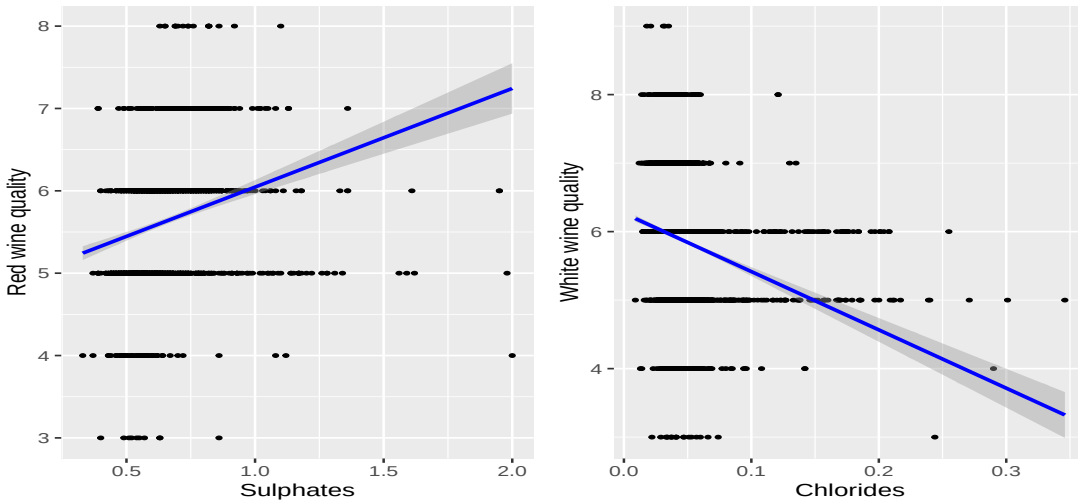

Figure 3: Examples of nonlinear relationship between sulphates and quality from red wine data set and between chlorides and quality from the white data set

`fit1`. We further try to investigate the relationship between the physicochemical variables for any relationship with wine quality. Notice that there is possibly a quadratic relationship between sulphates and quality, we fit a linear model including quadratic term of sulphates, indicating as model `fit2` in Table 3. Yet this model does not produce a lower AIC than `fit1`.

Model	Predictors Included	AIC
1	alc	3448
2	va, alc	3252
3	va, sul, alc	3209
4	va, tsd, sul, alc	3192
5	va, chl, tsd, sul, alc	3175
6	va, chl, tsd, ph, sul, alc	3163
7	va, chl, fsd, tsd, ph, sul, alc	3159
8	va, ca, chl, fsd, tsd, ph, sul, alc	3160

Table 2: Best subset selection results

Furthermore, we try to implement Generalized Additive Models for the model with three and four predictors as from best subset selection results. So, `fit3` and `fit4` are the

two attempts at GAMs with smoothing splines of degree of freedom 6. The two GAMs models result in lower AIC. Thus, `fit4` will be the chosen model. According to this model, volatile acidity, total sulfur dioxide, sulphates and alcohol play the roles in predicting red wine quality.

Model		Predictors Included	AIC
MLR	fit1	va, chl, fsd, tsd, ph, sul, alc	3259
MLR	fit2	va, alc, sul, sul^2	3160
GAM	fit3	va, sul, alc	3124
GAM	fit4	va, tsd, sul, alc	3104

Table 3: Best models for red wine data

3.2 White Wine Data Set

Using the same approach as for red wine samples (Gawel & Godden 2008), we come up with an MLR model that includes 7 predictors that has the lowest AIC out of the ones from best subset selections, as shown in Table 4. Then, fitting a GAM with some of these predictors gives a lower AIC as in Table 5.

Model	Predictors Included	AIC
1	alc	11685
2	va, alc	11373
3	va, sul, alc	11254
4	va, tsd, sul, alc	11175
5	va, chl, tsd, sul, alc	11139
6	va, chl, tsd, ph, sul, alc	11117
7	va, chl, fsd, tsd, ph, sul, alc	11108
8	va, ca, chl, fsd, tsd, ph, sul, alc	11113

Table 4: Best subset selection results

Model		Predictors Included	AIC
MLR	fit1	va, chl, fsd, tsd, ph, sul, alc	11108
GAM	fit2	va, rs, alc	11102
GAM	fit3	va, rs, fsd, alc	10807

Table 5: Best models for white wine data

On the basis of AIC, `fit7` would give the best prediction on wine quality. Thus, volatile acidity, residual sugar, free sulfur dioxide and alcohol are important in predicting the quality of white wine.

4 Results and Discussion

We have produced two models to predict wine quality of red and white wine separately. Multiple linear regression does not work out as well for prediction on the basis of AIC, perhaps due to the nonlinear functional relationships between the physicochemical variables and the wine quality. Generalized additive models have given better future prediction on the basis of AIC.

As a final note, this data set seems to be very nonlinear, thus nonlinear methods should be used to examine further models to improve prediction. Additionally, cross validation and other resampling methods could be used to access the error of the models to check for one with best prediction rate.

References

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